

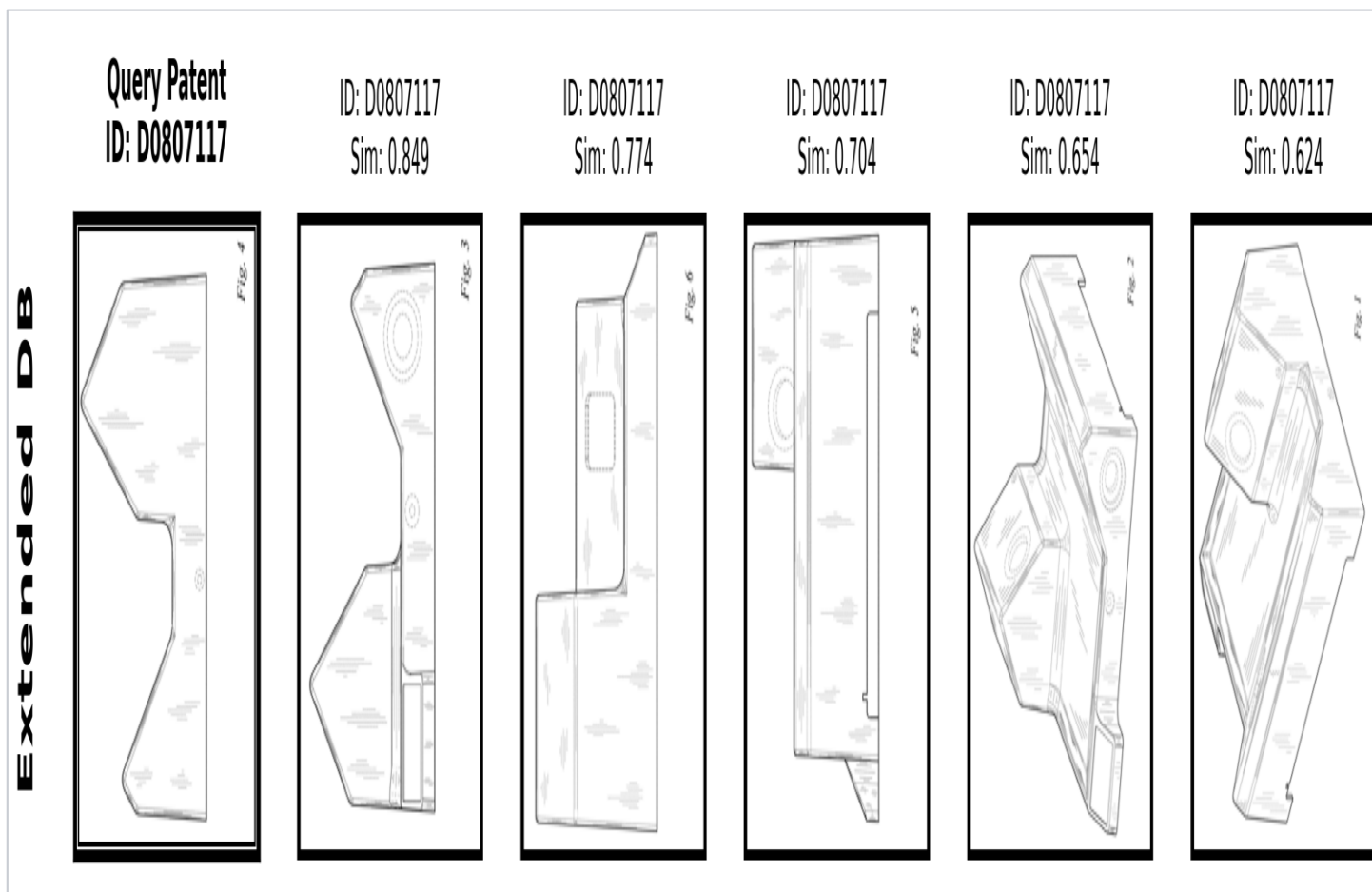
Design Patent Similarity with ArcFace Metric Learning

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Patent identity—not product category—is the right supervision for finding visually similar designs.

15-MINUTE RESEARCH TALK

Prior-art search is a near-duplicate problem



The retrieval target

Multiple views and minor drawing changes can belong to the same patented design.

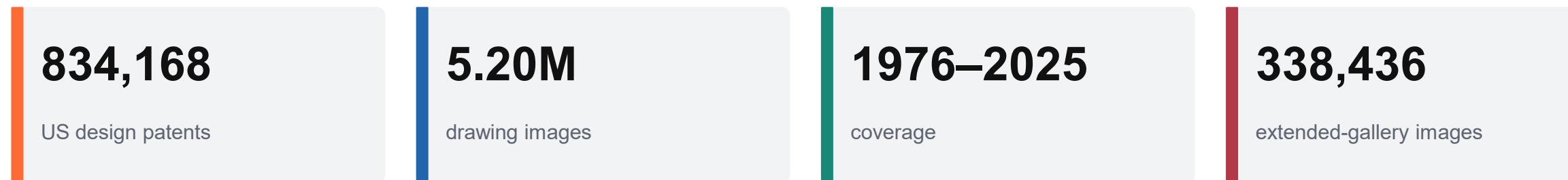
The failure mode

Category similarity is too coarse: two chairs can be legally and visually distinct.

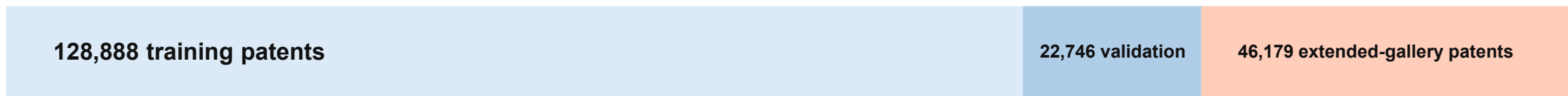
The research question

Can metric learning organize drawings around patent identity at national scale?

A national-scale corpus makes the task realistic



Train / validation / retrieval split



The extended gallery is 8.7× larger than the standard gallery—closer to production retrieval.

Patent IDs become the metric-learning signal



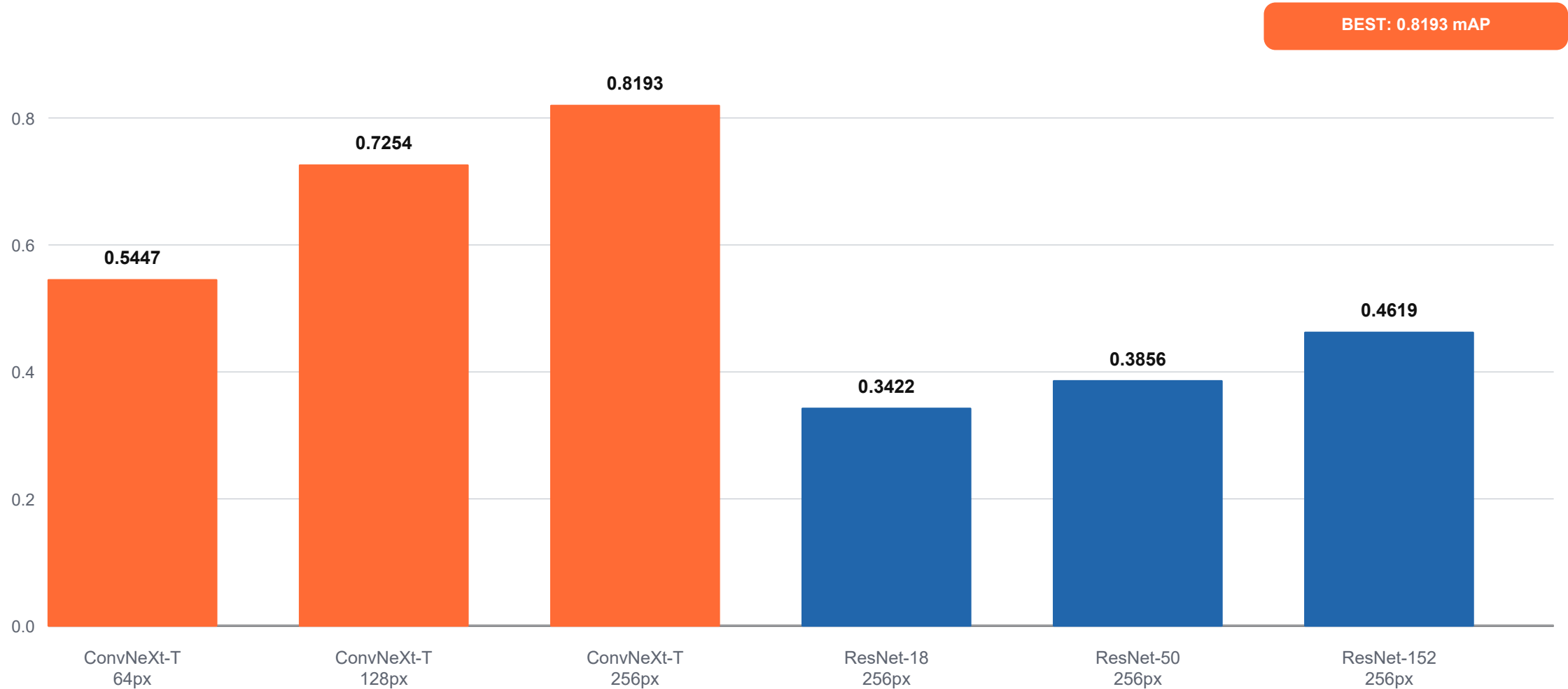
Training

Scale 30 · angular margin 0.50 · 30 epochs · effective batch ≈ 512

Retrieval

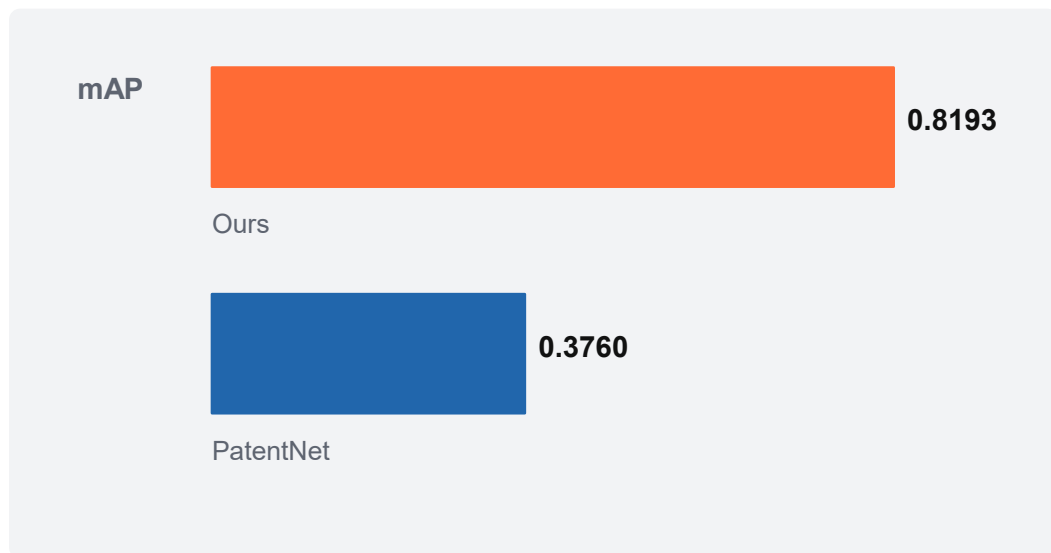
Cosine search, then optional query expansion or graph diffusion

Resolution and backbone capacity both matter



The gain survives a much larger search gallery

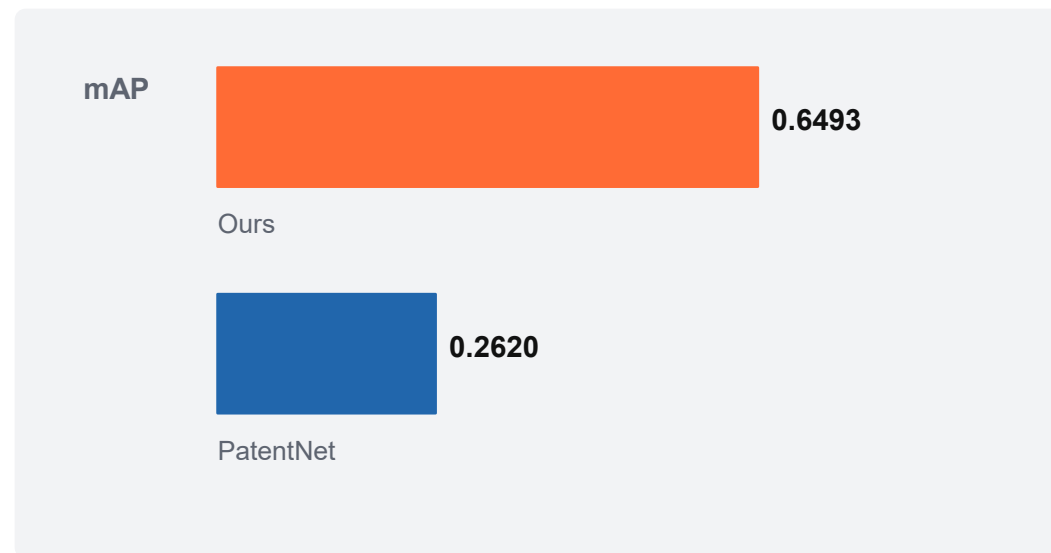
Standard Gallery



Acc@1 = 0.9312

Acc@20 = 0.9906

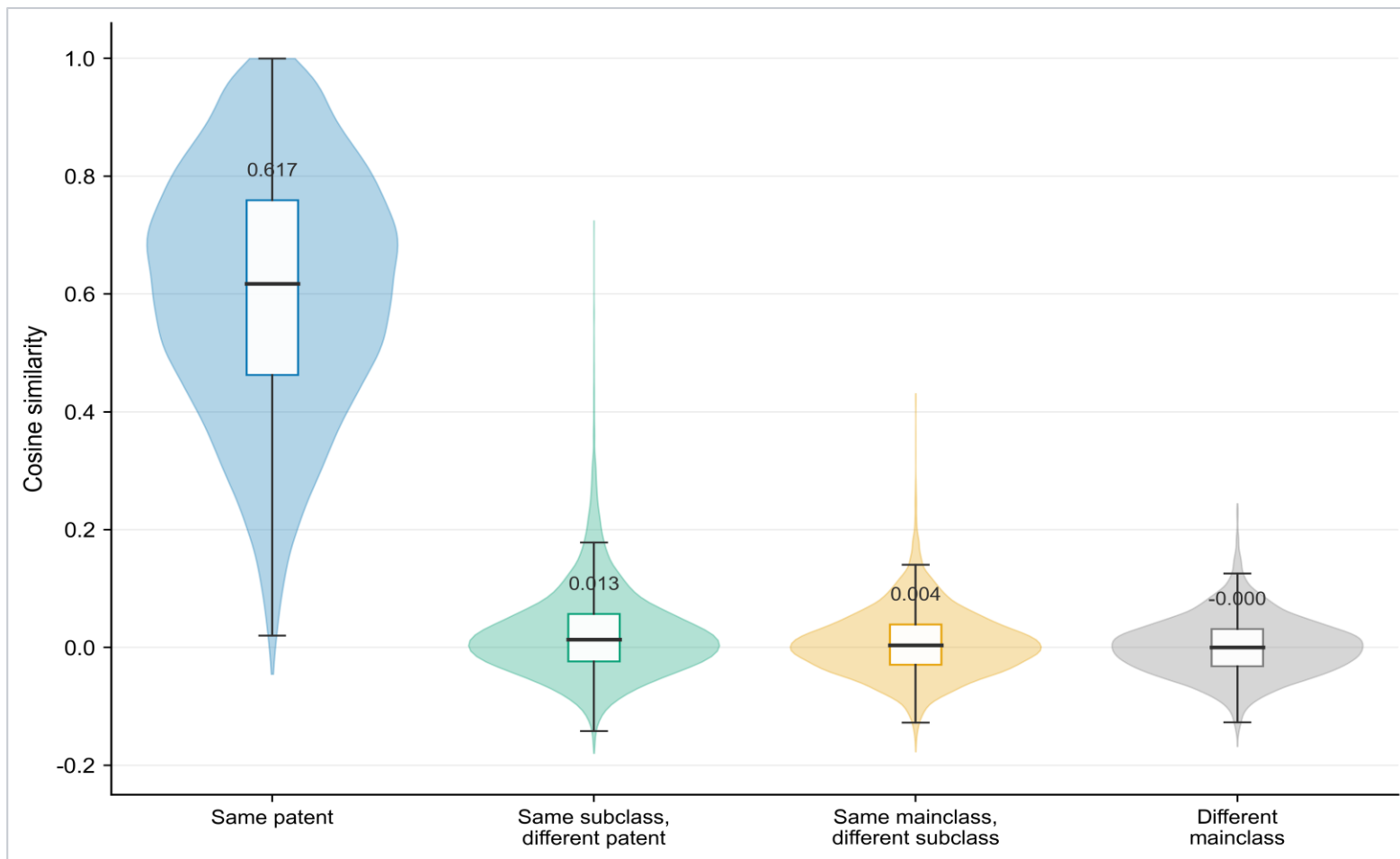
Extended Gallery



Acc@1 = 0.8088

Acc@20 = 0.9711

The embedding learns identity—not just taxonomy

**0.617**

median: same patent

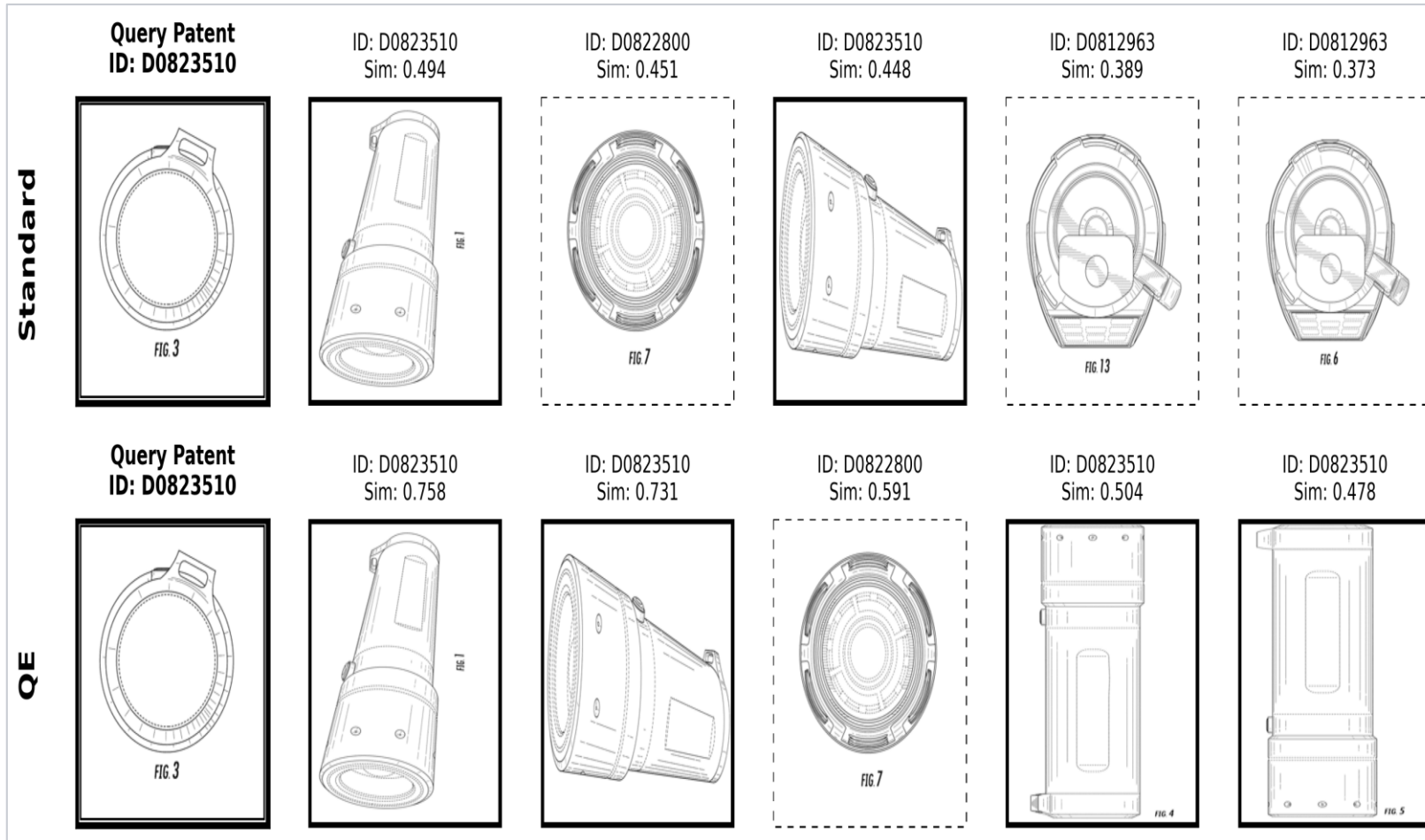
0.013

same subclass, other patent

≈0

different main class

Query expansion improves ranking—selectively



0.8193

baseline mAP

0.8417

query expansion mAP

Best setting

$k = 3 \cdot \alpha = 1$
Diffusion peaks at 0.8332.

A deployable retrieval layer—with clear limits

1 · Supervision

Patent identity turns multi-view drawings into strong positive pairs and avoids category-only shortcuts.

2 · Scale

High-resolution ConvNeXt features maintain useful precision in a 338k-image extended gallery.

3 · Next step

Hard negatives, domain filtering, and examiner-centered evaluation remain essential.

Bottom line: metric learning can narrow prior art; it does not replace legal judgment.